Universidad Distrital Francisco José de Caldas
School of Engineering – Computer Engineering Program

Data Analysis Programming – Semester 2026-II

Workshop No. 1

Hazard Definition, Source Validation, and Data Contract

Team 8 – Tsunami Monitoring

Lina Moreno	20231020133
Jaider Ávila	20231020200
Gabriela Martínez	20231020205

Professor: Eng. Carlos Andrés Sierra, M.Sc.

Repository: https://github.com/g8martinezsilva/tsunami-monitoring

November 7, 2026

Contents

1 Hazard Definition **2**
- 1.1 Phenomenon and Severity Measure . . . 2
- 1.2 Why Monitoring Matters . . . 2
- 1.3 Geographic and Temporal Scope . . . 2
- 1.4 Meaning of "Current Situation" . . . 2

2 Historical Sources **3**
- 2.1 NOAA/NCEI HazEL – Global Historical Tsunami Database . . . 3
- 2.2 USGS FDSN Event Web Service . . . 3

3 Incremental Sources **4**
- 3.1 USGS Real-Time GeoJSON Feed . . . 4
- 3.2 tsunami.gov Warning Centers . . . 4
- 3.3 Empirical Update Behavior . . . 4
- 3.4 Revised or Deleted Records and Watermark Implications . . . 4

4 Source Profiling and Data Quality **4**
- 4.1 Profiling Summary . . . 4
- 4.2 Data-Quality Problems . . . 5
- 4.3 Cross-Source Overlap . . . 5

5 Canonical Mapping **6**
- 5.1 Field Mapping to the Canonical `event` . . . 6
- 5.2 Domain Tables . . . 6
- 5.3 Unit Conversions and Rules . . . 6
- 5.4 Draft ER Diagram . . . 6

6 Ingestion Contract Draft **7**
- 6.1 Correctness After a Failed Run . . . 7

7 Project Definition **7**
- 7.1 Project Overview . . . 7
- 7.2 Functional User . . . 7
- 7.3 User Stories . . . 7
- 7.4 Indicator Proposal . . . 7
- 7.5 Risks and Mitigation . . . 8
- 7.6 Architecture Overview . . . 8

Conclusions and Open Questions **8**

A Real Response Snippets **9**

Executive Summary

[TODO: Five to eight lines: hazard, scope, the four sources, main findings of the profiling, and the proposed indicator.]

1 Hazard Definition

1.1 Phenomenon and Severity Measure

[TODO: What a tsunami is, how it is caused, how it is measured. Severity measure of the system (earthquake magnitude and its type, depth in km) and meaning of the units.]

1.2 Why Monitoring Matters

[TODO: Motivation. State that the system supports monitoring and correlation and does **not** predict tsunamis.]

1.3 Geographic and Temporal Scope

The project covers the Pacific basin, including Alaska and Hawaii, during 2010–2025, with earthquakes of magnitude $M \geq 6$. NOAA HazEL regions 80–89 and a USGS rectangle (latitude −60 to 65, longitude 100 to 300) provide geographic filters; the USGS historical sample covers 2010–2011 to meet the workshop's 200-record requirement. The two real-time feeds are profiled in full, with Pacific events identified by coordinates; tsunami.gov covers the Pacific, Alaska, the US and the Caribbean.

Source	Coverage	Fits the scope?
NOAA HazEL	Global	Yes, regions 80–89
USGS FDSN	Global	Yes, Pacific rectangle
USGS real-time feeds	Global	Yes, in-scope events identified by coordinates
tsunami.gov	Pacific, Alaska, US, Caribbean	Yes, in-scope events identified by coordinates

1.4 Meaning of "Current Situation"

[TODO: For example: earthquakes of the last 24 hours and 7 days, and the latest tsunami.gov message.]

2 Historical Sources

2.1 NOAA/NCEI HazEL – Global Historical Tsunami Database

Owner	[TODO: NOAA National Centers for Environmental Information (NCEI)]
URL	https://www.ngdc.noaa.gov/hazel/view/swagger
Access method	REST API, JSON
Authentication	None required
Rate limits	[TODO: verify and cite]
License and attribution	[TODO: verify and cite]
Fields used	[TODO: list, e.g. year, month, day, latitude, longitude, causeCode, eventValidity, eqMagnitude]
Time coverage	[TODO: from the catalog start to the present]
Update policy	[TODO:]

Evidence. [TODO: Real response snippet (see Appendix A) and the file under `workshop_1/samples/`.]

Volume and partitioning. [TODO: Records in scope (2010–2025, regions 80–89), size, pagination of 200 records per page, and how the bulk load will be partitioned.]

2.2 USGS FDSN Event Web Service

Owner	U.S. Geological Survey, Earthquake Hazards Program
URL	https://earthquake.usgs.gov/fdsnws/event/1/
Access method	REST API, CSV (also GeoJSON and QuakeML)
Authentication	None required
Rate limits	[TODO: maximum 20,000 events per request; cite documentation]
License and attribution	[TODO: verify and cite]
Fields used	[TODO: id, time, latitude, longitude, depth, mag, magType, status, updated]
Time coverage	[TODO:]
Update policy	[TODO: reviewed events are revised later; see Section 3]

Evidence. [TODO:]

Volume and partitioning. [TODO: Estimate for 2010–2025 and the decision to load one year at a time.]

3 Incremental Sources

3.1 USGS Real-Time GeoJSON Feed

URL	https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php
Feed used	Magnitude 4.5+, past 30 days
Format	GeoJSON
Update frequency	[TODO: the server sets a cache of 60 seconds]
Change detection	[TODO: `updated` field, stable `id`, Last-Modified header]

3.2 tsunami.gov Warning Centers

URL (NTWC)	https://www.tsunami.gov/events/xml/PAAQAtom.xml
URL (PTWC)	https://www.tsunami.gov/events/xml/PHEBAtom.xml
Format	Atom feed; each entry links to a CAP message
Content	[TODO: one latest message per feed; most messages are information statements]
Change detection	[TODO: entry id, `updated`, Last-Modified header]

3.3 Empirical Update Behavior

Two snapshots were taken on 7 November 2026 at 05:04 and 14:03 UTC (about 9 hours apart).

	USGS feed	tsunami.gov (NTWC, PTWC)
First snapshot (records)	714	1 per feed
Second snapshot (records)	726	1 per feed
New	22	0
Modified	25	0
Removed	10	0

Table 1: Changes between the two snapshots.

[TODO: Explain: all 10 removed events left the rolling 30-day window (none was deleted); fields that changed in the modified events (time, place, magnitude, station counts); event ids stayed stable.]

3.4 Revised or Deleted Records and Watermark Implications

[TODO: Watermark on `updated`, upsert on (source, id), content hash to separate real changes from timestamp-only updates, overlap window.]

4 Source Profiling and Data Quality

4.1 Profiling Summary

[TODO: Rows, null percentages, ranges, duplicated identifiers, timestamp format and time zone, coordinate validity and units for each source. Reference the notebook `notebooks/01_source_profiling.ipynb`.]

Source	Records	Duplicated ids	Notes
NOAA HazEL	[TODO:]	[TODO:]	[TODO:]
USGS FDSN	[TODO:]	[TODO:]	[TODO:]
USGS real-time	[TODO:]	[TODO:]	[TODO:]
tsunami.gov	[TODO:]	[TODO:]	[TODO:]

4.2 Data-Quality Problems

At least five, each with an example record.

#	Source	Problem	Example record
1	NOAA HazEL	Official warning status missing	[TODO:]
2	NOAA HazEL	Erroneous or doubtful events	[TODO:]
3	NOAA HazEL	Time of day or coordinates missing	[TODO:]
4	USGS FDSN	Columns that are always empty	[TODO:]
5	tsunami.gov	Location as free text; depth units differ	[TODO:]

4.3 Cross-Source Overlap

At least five candidate pairs describing the same real-world event.

#	Source A id	Source B id	Δt	Distance	Magnitude	What differs
1	PAAQ-1-to0j1g	nc75462637	-1.3 s	8.7 km	4.5 Ml / 4.96 mw	magnitude type, depth, location text
2	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
3	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
4	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]
5	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]	[TODO:]

5 Canonical Mapping

5.1 Field Mapping to the Canonical `event`

Canonical field	NOAA HazEL	USGS FDSN / feed	tsunami.gov (CAP)	Rule
`event_id`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`source`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`event_type`	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`timestamp` (UTC)	[TODO:]	[TODO:]	[TODO:]	[TODO:]
`latitude`	[TODO:]	[TODO:]	[TODO:]	WGS84
`longitude`	[TODO:]	[TODO:]	[TODO:]	WGS84, -180 to 180
`severity`	[TODO:]	[TODO:]	[TODO:]	[TODO: value and unit/type]
`raw_reference`	[TODO:]	[TODO:]	[TODO:]	id or URL plus content hash

5.2 Domain Tables

[TODO: Proposed tables (for example `earthquake`, `tsunami_event`, `tsunami_alert`) with their attributes.]

5.3 Unit Conversions and Rules

[TODO: Timestamps (UTC), coordinates (WGS84), longitude normalization, depth units, magnitude types and severity scale.]

5.4 Draft ER Diagram



Figure 1: First draft of the entity-relationship diagram.

6 Ingestion Contract Draft

Source	Historical strategy	Incremental strategy	Duplicate key	Overlap window
NOAA HazEL	[TODO:]	[TODO:]	[TODO:]	[TODO:]
USGS FDSN	[TODO:]	[TODO:]	[TODO:]	[TODO:]
USGS real-time feed	–	[TODO:]	[TODO:]	[TODO:]
tsunami.gov	–	[TODO:]	[TODO:]	[TODO:]

6.1 Correctness After a Failed Run

[TODO: The watermark advances only after a successful commit; idempotent writes; what happens to a partially loaded batch.]

7 Project Definition

7.1 Project Overview

[TODO: Title, motivation, goal (data-engineering and analytical), scope, explicit exclusions and constraints.]

7.2 Functional User

[TODO: Role, profile, decisions supported, and why the dashboard is useful.]

7.3 User Stories

ID	Type	Story	MoSCoW
US1	Monitoring	As a [TODO: user], I want to [TODO: action] so that [TODO: benefit].	[TODO:]
US2	Monitoring	[TODO:]	[TODO:]
US3	Analysis	[TODO:]	[TODO:]
US4	Analysis	[TODO:]	[TODO:]
US5	Trust	[TODO:]	[TODO:]
US6	Trust	[TODO:]	[TODO:]

7.4 Indicator Proposal

[TODO: Name, inputs (magnitude, depth, location, historical tsunami association, official alert), rule, output scale, how it will be validated against historical events. State clearly that it is a monitoring aid and not a prediction, and list its limitations.]

7.5 Risks and Mitigation

Risk	Impact	Mitigation
Source outage	[TODO:]	[TODO:]
Rate limits	[TODO:]	[TODO:]
Schema change	[TODO:]	[TODO:]
License and attribution	[TODO:]	[TODO:]
Ambiguous identifiers	[TODO:]	[TODO:]

7.6 Architecture Overview

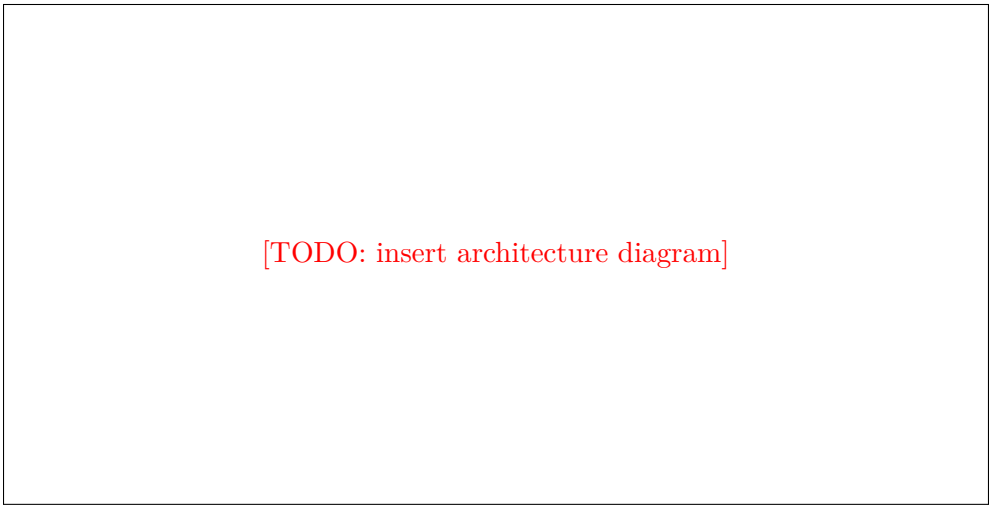


Figure 2: High-level architecture adapted to tsunami monitoring.

Conclusions and Open Questions

[TODO: Main decisions taken in this workshop and what remains open for Workshop 2.]

References

[1] NOAA National Centers for Environmental Information. *Hazard Event Database (HazEL), Global Historical Tsunami Database*. https://www.ngdc.noaa.gov/hazel/view/swagger. Accessed [TODO: date].

[2] U.S. Geological Survey. *API Documentation – Earthquake Catalog*. https://earthquake.usgs.gov/fdsnws/event/1/. Accessed [TODO: date].

[3] U.S. Geological Survey. *Real-time Notifications, Feeds, and Web Services (GeoJSON summary feed)*. https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php. Accessed [TODO: date].

[4] NOAA National Weather Service, Tsunami Warning Centers. *Tsunami event feeds*. https://www.tsunami.gov/. Accessed [TODO: date].

A Real Response Snippets

[TODO: Short excerpts of the unmodified responses, one per source, with the file name in `workshop_1/samples/`.]

```
(paste a short excerpt here)
```